

# Chapter 18, 19 & 21: Electrical, Thermal & Optical Properties of Materials

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Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

**Course:** Materials Science (MIAE 221)

**Textbook:**

*Materials Science and Engineering: An Introduction*

(10th Edition) by William D. Callister, Jr. & David G. Rethwisch — Chapters 18, 19 & 21

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## 1. Executive Overview & First-Principles Philosophy

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In previous chapters, we focused primarily on how materials respond to mechanical forces (stress, strain, dislocation slip). In modern high-technology engineering, materials must simultaneously fulfill critical **functional roles**:

- A microelectronic processor relies on millions of transistors operating at gigahertz switching speeds governed by **electronic bandgaps and semiconductor charge carrier mobilities**.
- A spacecraft thermal protection tile (e.g., Space Shuttle silica tiles) must withstand 1200°C re-entry plasma while keeping the underlying aluminum fuselage cold, governed by **thermal conductivity, thermal expansion, and thermal shock resistance**.
- Optical fiber communications transmit terabits of data per second across oceans via laser pulses through ultra-pure silica glass, governed by **refractive indices, photon absorption bandgaps, and Rayleigh scattering**.

All of these physical phenomena are direct manifestations of **electronic energy band structures and quantized atomic vibrations (phonons)**.

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## 2. Electrical Properties: Band Theory & Semiconductors (Callister Chapter 18)

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### 2.1 Ohm's Law & Electrical Conductivity

The macroscopic electrical resistance  $R$  of a conductor of length  $l$  and uniform cross-sectional area  $A$  is described by Ohm's Law:

$$V = IR \iff R = \frac{V}{I}$$

Because resistance depends on specimen geometry, materials engineers normalize by dimensions to define intrinsic **Electrical Resistivity** ( $\rho$ ):

$$\rho = \frac{RA}{l} \quad [\Omega \cdot \text{m}]$$

The inverse of resistivity is **Electrical Conductivity** ( $\sigma$ ):

$$\sigma = \frac{1}{\rho} = \frac{l}{RA} \quad [(\Omega \cdot \text{m})^{-1} \text{ or S/m}]$$

At the microscopic scale:

$$J = \sigma \mathcal{E}$$

where  $J = I/A$  is current density ( $\text{A/m}^2$ ) and  $\mathcal{E} = V/l$  is electric field intensity ( $\text{V/m}$ ).

### The Microscopic Charge Carrier Equation:

Electrical conduction occurs via the drift of charged particles under an applied electric field:

$$\sigma = n|q|\mu_e$$

where:

- $n$  = number of free charge carriers per unit volume ( $\text{m}^{-3}$ ).
- $q$  = carrier charge (for electrons,  $e = 1.602 \times 10^{-19} \text{ C}$ ).
- $\mu_e$  = **carrier mobility** ( $\text{m}^2/(\text{V} \cdot \text{s})$ ), defined as drift velocity per unit electric field:  $v_d = \mu_e \mathcal{E}$ .

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## 2.2 Electron Energy Band Structures in Solids

In isolated atoms, electrons occupy discrete atomic quantum energy levels ( $1s, 2s, 2p, \dots$ ). When  $N$  atoms assemble into a solid crystal, the Pauli Exclusion Principle splits each atomic energy level into  $N$  closely spaced, overlapping quantum states, forming continuous **Electron Energy Bands**:

- **Valence Band**: The highest occupied energy band filled with valence electrons at 0 K.
- **Conduction Band**: The lowest unoccupied (or partially filled) energy band into which electrons can be promoted to move freely under an electric field.
- **Fermi Energy** ( $E_F$ ): The highest energy state occupied by electrons at absolute zero temperature (0 K).
- **Bandgap** ( $E_g$ ): An energetically forbidden gap of energy states separating the top of the valence band from the bottom of the conduction band.

**Figure 18.4** The various possible electron band structures in solids at 0 K. (a) The electron band structure found in metals such as copper, in which there are available electron states above and adjacent to filled states, in the same band. (b) The electron band structure of metals such as magnesium, in which there is an overlap of filled and empty outer bands. (c) The electron band structure characteristic of insulators; the filled valence band is separated from the empty conduction band by a relatively large band gap ( $>2$  eV). (d) The electron band structure found in the semiconductors, which is the same as for insulators except that the band gap is relatively narrow ( $<2$  eV).

The number of states within each band is equal to the total of all states contributed by the  $N$  atoms. For example, an  $s$  band consists of  $N$  states and a  $p$  band of  $3N$  states. With regard to occupancy, each energy state may accommodate two electrons that must have oppositely directed spins. Furthermore, bands contain the electrons that resided in the corresponding levels of the isolated atoms; for example, a  $4s$  energy band in the solid contains those isolated atoms'  $4s$  electrons. Of course, there are empty bands and, possibly, bands that are only partially filled.

The electrical properties of a solid material are a consequence of its electron band structure—that is, the arrangement of the outermost electron bands and the way in which they are filled with electrons.

Four different types of band structures are possible at 0 K. In the first (Figure 18.4a), one outermost band is only partially filled with electrons. The energy corresponding to the highest filled state at 0 K is called the **Fermi energy**  $E_F$ , as indicated. This energy band structure is typified by some metals, in particular those that have a single  $s$  valence electron (e.g., copper). Each copper atom has one  $4s$  electron; however, for a solid composed of  $N$  atoms, the  $4s$  band is capable of accommodating  $2N$  electrons. Thus, only half of the available electron positions within this  $4s$  band are filled.

For the second band structure, also found in metals (Figure 18.4b), there is an overlap of an empty band and a filled band. Magnesium has this band structure. Each

**Fermi energy**

Figure 18.4: Schematic representations of the four primary electron energy band structures: (a) Partially filled conduction band (Metals like Cu, Na), (b) Overlapping valence and conduction bands (Metals like Mg, Zn), (c) Wide bandgap Insulator ( $E_g > 2$  eV), and (d) Narrow bandgap Semiconductor ( $E_g < 2$  eV) — from Callister & Rethwisch 10th Ed. (Fig. 18.4).

## The Four Classification Regimes:

### 1. Metals (Conductors) ( $\sigma \approx 10^7$ ( $\Omega \cdot \text{m}$ ) $^{-1}$ ):

- Possess either a **partially filled band** (Fig. 18.4a) or **overlapping valence and conduction bands** (Fig. 18.4b).
- The Fermi level  $E_F$  lies inside a continuous band. Countless vacant energy states exist immediately adjacent to  $E_F$ . Electrons need almost zero thermal energy to jump into empty states and conduct electricity.

### 2. Insulators ( $\sigma \approx 10^{-20} - 10^{-10}$ ( $\Omega \cdot \text{m}$ ) $^{-1}$ ):

- Possess a completely filled valence band, a completely empty conduction band, and a **wide bandgap** ( $E_g > 2.0$  eV) (e.g., Diamond  $E_g = 5.5$  eV, Alumina  $E_g = 9.0$  eV).
- At room temperature, thermal kinetic energy ( $k_B T \approx 0.026$  eV) is orders of magnitude smaller than  $E_g$ . Essentially zero electrons are promoted to the conduction band  $\Rightarrow$  electrical conductivity is negligible.

### 3. Semiconductors ( $\sigma \approx 10^{-6} - 10^4$ ( $\Omega \cdot \text{m}$ ) $^{-1}$ ):

- Possess a **narrow bandgap** ( $E_g < 2.0$  eV) (e.g., Silicon  $E_g = 1.1$  eV, Germanium  $E_g = 0.67$  eV).
- Modest thermal energy promotes a small but significant number of electrons across  $E_g$  into the conduction band.

## 2.3 Intrinsic vs. Extrinsic Semiconductors

### A. Intrinsic (Pure) Semiconductors

In an ultra-pure semiconductor (e.g., pure Silicon), electrical conduction relies solely on thermally excited charge carriers:

- When an electron jumps across  $E_g$  into the conduction band, it leaves behind an empty electron state in the valence band—a positively charged entity called a **Hole** ( $h^\bullet$ ).
- Both electrons ( $n$ ) and holes ( $p$ ) drift under an applied electric field, contributing to total conductivity:

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

- In intrinsic material, electrons and holes are created in identical pairs:

$$n = p = n_i$$

$$\sigma = n_i|e|(\mu_e + \mu_h)$$

- **Temperature Dependence:** Intrinsic carrier concentration obeys an exponential Arrhenius relation:

$$n_i \propto \exp\left(-\frac{E_g}{2k_B T}\right)$$

-  **Crucial Contrast with Metals:**
  - In **metals**, increasing temperature increases lattice vibrations (phonons), which scatter electrons and **decrease conductivity** ( $\sigma$  drops as  $T$  rises).
  - In **intrinsic semiconductors**, increasing temperature exponentially multiplies the number of free carriers ( $n_i$ ), causing **conductivity to surge exponentially** as temperature rises!

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### B. Extrinsic (Doped) Semiconductors

Commercial semiconductor devices are doped by intentionally introducing tiny concentrations (parts-per-million) of specific impurity atoms:

## Extrinsic Semiconductors

### n-TYPE SEMICONDUCTOR

- Doped with Group V donor elements (P, As, Sb in Si) elements (B, Al, Ga in Si)
- 5 valence electrons: 4 form covalent bonds, 1 is extra missing an electron
- Donor energy level  $E_d$  sits just below conduction band ( $\sim 0.05$  eV)
- Donates conduction electrons:  $n \gg p$  holes:  $p \gg n$
- Majority carriers: ELECTRONS
- $\sigma \approx n |e| \mu_e \approx N_d |e| \mu_e$

### p-TYPE SEMICONDUCTOR

- Doped with Group III acceptor
- 3 valence electrons: 1 bond is missing an electron
- Acceptor level  $E_a$  sits just above valence band ( $\sim 0.05$  eV)
- Captures electrons, creating excess holes
- Majority carriers: HOLES
- $\sigma \approx p |e| \mu_h \approx N_a |e| \mu_h$

## C. The Three Temperature Regimes of Extrinsic Semiconductors:

1. **Freeze-Out Regime** ( $T < 100$  K): Thermal energy is too low to ionize dopant atoms ( $k_B T < 0.05$  eV). Carriers remain trapped on impurity atoms; conductivity is low.
2. **Extrinsic / Saturation Regime** ( $100 \text{ K} < T < 450 \text{ K}$  — **Normal Operating Window**):
  - Thermal energy is sufficient to ionize virtually **all donor or acceptor atoms**:

$$n \approx N_d \quad (\text{for } n\text{-type}) \quad \text{or} \quad p \approx N_a \quad (\text{for } p\text{-type})$$

- Carrier concentration remains constant. In this regime, conductivity decreases slightly as temperature rises because increased lattice phonon vibrations reduce electron mobility ( $\mu \propto T^{-3/2}$ ).
3. **Intrinsic Regime** ( $T > 450$  K):
    - Thermal excitation of electron-hole pairs across the full bandgap ( $E_g = 1.1$  eV) completely overwhelms the dopant concentration ( $n_i \gg N_d$ ).
    - The semiconductor loses its extrinsic  $n$ -type or  $p$ -type character and behaves as an intrinsic semiconductor, causing integrated circuit transistors to short-circuit and fail!

## 3. Thermal Properties of Materials (Callister Chapter 19)

### 3.1 Heat Capacity ( $C_v, C_p$ ) & Phonons

**Heat capacity** represents a material's ability to absorb thermal energy from its surroundings:

$$C = \frac{dQ}{dT}$$

- At the atomic scale, thermal energy is stored primarily as **atomic lattice vibrations**. In a solid crystal, vibrations are coupled into traveling elastic waves called **Phonons** (quantized vibrational waves traveling at the speed of sound).
- **The Debye Temperature** ( $\theta_D$ ):

- At cryogenic temperatures ( $T < \theta_D$ ), heat capacity increases rapidly with the cube of temperature:

$$C_v \propto T^3 \quad (\text{Debye } T^3 \text{ Law})$$

- At temperatures well above  $\theta_D$ , heat capacity levels off to the constant classical **Dulong-Petit limit**:

$$C_v \approx 3R \approx 3(8.314) \approx 25 \text{ J}/(\text{mol} \cdot \text{K})$$

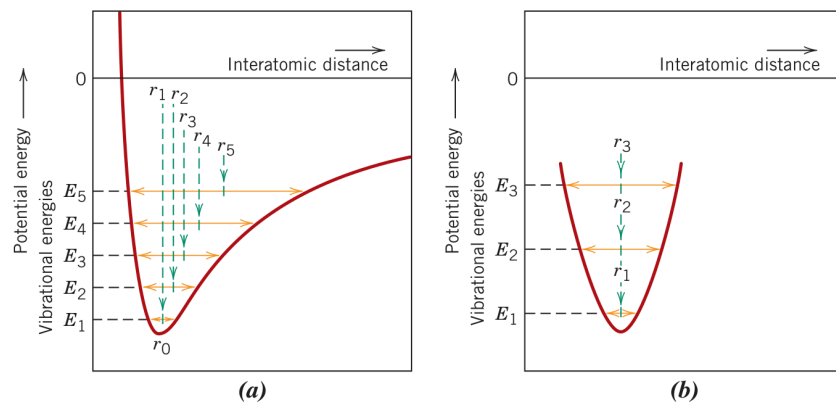
### 3.2 Thermal Expansion & Potential Energy Well Asymmetry

When a solid is heated, its linear dimensions expand proportionally:

$$\frac{\Delta l}{l_0} = \alpha_l \Delta T \iff \Delta l = \alpha_l l_0 (T_f - T_0)$$

where  $\alpha_l$  is the **Linear Coefficient of Thermal Expansion** ( $\text{K}^{-1}$  or  $^{\circ}\text{C}^{-1}$ ).

average distance between the atoms. This phenomenon can best be understood by consulting the potential energy–versus–interatomic spacing curve for a solid material introduced previously (Figure 2.10b) and reproduced in Figure 19.3a. The curve is in



**Figure 19.3** (a) Plot of potential energy versus interatomic distance, demonstrating the increase in interatomic separation with rising temperature. With heating, the interatomic separation increases from  $r_0$  to  $r_1$  to  $r_2$ , and so on. (b) For a symmetric potential energy–versus–interatomic distance curve, there is no increase in interatomic separation with rising temperature (i.e.,  $r_1 = r_2 = r_3$ ).

(Adapted from R. M. Rose, L. A. Shepard, and J. Wulff, *The Structure and Properties of Materials*, Vol. IV, *Electronic Properties*, John Wiley & Sons, 1966. Reproduced with permission of Robert M. Rose.)

Figure 19.3: Physical origin of thermal expansion: (a) Symmetric potential energy well yields zero net thermal expansion, whereas (b) Real asymmetric (anharmonic) potential energy well causes the average atomic separation distance to shift outward ( $\bar{r}_T > r_0$ ) as thermal energy rises — from Callister & Rethwisch 10th Ed. (Fig. 19.3).

- Deep, narrow wells** (Diamond, Alumina, Tungsten) have high bonding energies, high symmetry, and very low thermal expansion ( $\alpha_l \approx 1 - 8 \times 10^{-6} \text{ K}^{-1}$ ).
- Shallow, wide wells** (Polymers) are highly asymmetric, yielding massive thermal expansion ( $\alpha_l \approx 50 - 200 \times 10^{-6} \text{ K}^{-1}$ ).

### 3.3 Thermal Conductivity ( $k$ ) & The Wiedemann-Franz Law

Thermal conduction is the transport of heat from high-temperature to low-temperature regions:

$$q = -k \frac{dT}{dx}$$

where  $q$  is heat flux ( $\text{W}/\text{m}^2$ ) and  $k$  is **Thermal Conductivity** ( $\text{W}/(\text{m} \cdot \text{K})$ ).

Heat is transported through solids via **two concurrent mechanisms**:

$$k = k_e + k_l$$

1. **Electronic Thermal Conduction ( $k_e$ )**: Carried by the kinetic energy of highly mobile free conduction electrons.
2. **Lattice Vibration Conduction ( $k_l$ )**: Carried by traveling phonon waves.

- **In Metals**: Free electrons are abundant  $\implies k_e \gg k_l$ . Electrons conduct heat and electricity via the exact same physical particles!
  - **The Wiedemann-Franz Law**: For pure metals, the ratio of thermal conductivity to electrical conductivity is directly proportional to absolute temperature:

$$\frac{k}{\sigma T} = L \approx 2.44 \times 10^{-8} \Omega \cdot \text{W}/\text{K}^2 \quad (\text{Lorentz Number } L)$$

- **In Ceramics & Polymers**: Essentially zero free electrons exist  $\implies k_e = 0$ . Heat is conducted **purely by phonons** ( $k = k_l$ ). Because phonons scatter heavily at grain boundaries, voids, and disordered polymer chains, nonmetals generally exhibit low thermal conductivities ( $k \approx 0.1 - 2 \text{ W}/(\text{m} \cdot \text{K})$ ).

- *The Diamond Exception*

: Pure single-crystal diamond has no grain boundaries and exceptionally stiff covalent bonds, yielding the highest thermal conductivity of any known bulk solid ( $k \approx 2000 \text{ W}/(\text{m} \cdot \text{K})$ ) purely through ultra-fast phonon transport!

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### 3.4 Thermal Stresses & Thermal Shock Resistance (TSR)

When a material is heated or cooled while mechanically constrained from expanding or contracting, **thermal stresses** ( $\sigma_{\text{thermal}}$ ) develop:

$$\sigma = -E\alpha_l \Delta T$$

If the material is cooled rapidly ( $\Delta T < 0$ ), tensile stresses develop on the surface. In brittle ceramics with low fracture toughness, these thermal tensile stresses can cause catastrophic cracking and shattering—a failure mode termed **Thermal Shock**.

**The Thermal Shock Resistance (TSR)** parameter measures a brittle material's ability to endure rapid temperature changes without fracture:

$$\text{TSR} \cong \frac{\sigma_f k}{E \alpha_l}$$

To maximize thermal shock resistance, an engineer must select a material with:

- High fracture strength ( $\sigma_f$ )
- High thermal conductivity ( $k$ ) (rapidly dissipates temperature gradients)
- Low Young's modulus ( $E$ ) (flexible, compliant)
- **Very low thermal expansion coefficient ( $\alpha_l$ )** (minimizes thermal strain!)
- *Engineering Example*

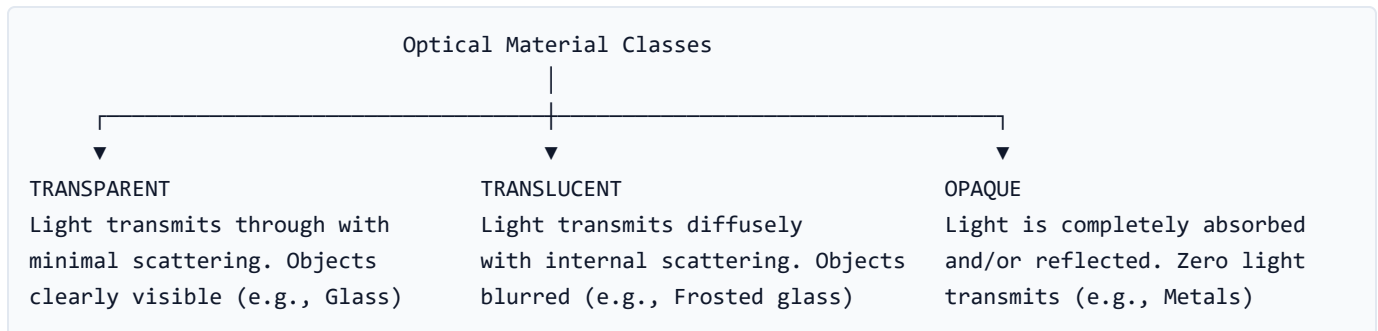
: Why does borosilicate Pyrex glass endure oven-to-sink thermal shocks while ordinary soda-lime window glass shatters? Because Pyrex has an exceptionally low thermal expansion coefficient ( $\alpha_{\text{Pyrex}} = 3.3 \times 10^{-6} \text{ K}^{-1}$  vs.  $\alpha_{\text{soda-lime}} = 9.0 \times 10^{-6} \text{ K}^{-1}$ ), giving it roughly **three times higher TSR!**

## 4. Optical Properties of Materials (Callister Chapter 21)

When electromagnetic radiation (light) strikes a solid, the incident light intensity  $I_0$  is partitioned among three physical processes:

$$I_0 = I_R + I_A + I_T \implies R + A + T = 1$$

where  $R = I_R/I_0$  is **reflectivity**,  $A = I_A/I_0$  is **absorptivity**, and  $T = I_T/I_0$  is **transmissivity**.



### 4.1 Optical Properties of Metals: High Reflectivity & Opacity

- Metals are completely **opaque** across the entire visible spectrum ( $0.4 \mu\text{m} \leq \lambda \leq 0.7 \mu\text{m}$ ) within a skin depth of only a few nanometers.
- *Physical Mechanism*

: Because metals have continuous empty electron energy states immediately above the Fermi level, valence electrons absorb incident visible photons of all wavelengths ( $h\nu$ ). The excited electrons immediately re-radiate electromagnetic waves of the identical frequency in the backward direction as **reflected light**, giving polished metals their characteristic shiny metallic luster ( $R \approx 90 - 95$ ).

## 4.2 Optical Properties of Nonmetals: The Bandgap Criterion

In non-metallic materials (ceramics and polymers), optical absorption occurs via **interband electron transitions**: an incident photon can be absorbed only if its energy  $E = h\nu$  is sufficient to excite an electron from the valence band across the bandgap into the conduction band:

$$h\nu \geq E_g$$

### Optical Properties of Metals

Consider the electron energy band schemes for metals as illustrated in Figures 18.4a and 18.4b; in both cases, a high-energy band is only partially filled with electrons. Metals are opaque because the incident radiation having frequencies within the visible range excites electrons into unoccupied energy states above the Fermi energy, as demonstrated in Figure 21.4a; as a consequence, the incident radiation is absorbed, in accordance with Equation 21.6. Total absorption is within a very thin outer layer, usually less than  $0.1\ \mu\text{m}$ ; thus only metallic films thinner than  $0.1\ \mu\text{m}$  are capable of transmitting visible light.

All frequencies of visible light are absorbed by metals because of the continuously available empty electron states, which permit electron transitions as in Figure 21.4a. In fact, metals are opaque to all electromagnetic radiation on the low end of the frequency

**Figure 21.4** (a) Schematic representation of the mechanism of photon absorption for metallic materials in which an electron is excited into a higher-energy unoccupied state. The change in energy of the electron  $\Delta E$  is equal to the energy of the photon. (b) Reemission of a photon of light by the direct transition of an electron from a high to a low energy state.

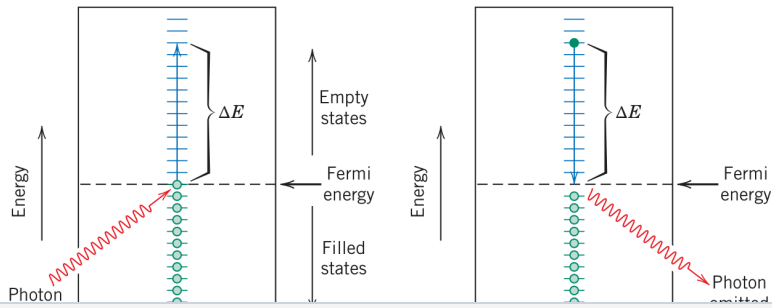


Figure 21.4: Mechanism of optical absorption in nonmetallic solids: (a) An incident photon with energy  $h\nu \geq E_g$  excites an electron across the bandgap. (b) If  $h\nu < E_g$ , the photon cannot be absorbed and passes through uninhibited — from Callister & Rethwisch 10th Ed. (Fig. 21.4).

### Mathematical Relationship Between Wavelength and Energy:

Photon energy is related to wavelength  $\lambda$  by:

$$E = h\nu = \frac{hc}{\lambda}$$

where  $h = 6.626 \times 10^{-34}\ \text{J} \cdot \text{s} = 4.136 \times 10^{-15}\ \text{eV} \cdot \text{s}$ ,  $c = 3.0 \times 10^8\ \text{m/s}$ . In practical engineering units:

$$E(\text{eV}) = \frac{1.24}{\lambda(\mu\text{m})} \iff \lambda(\mu\text{m}) = \frac{1.24}{E(\text{eV})}$$

The visible light spectrum spans photon energies from:

- Red edge ( $\lambda = 0.7\ \mu\text{m}$ ):  $E_{\min} \approx \frac{1.24}{0.7} \approx 1.8\ \text{eV}$
- Violet edge ( $\lambda = 0.4\ \mu\text{m}$ ):  $E_{\max} \approx \frac{1.24}{0.4} \approx 3.1\ \text{eV}$

### The 3 Optical Regimes for Nonmetals:

#### 1. Wide Bandgap Insulators ( $E_g > 3.1$ eV):

- Visible photons (1.8 – 3.1 eV) have **insufficient energy to excite electrons across  $E_g$** .
- Photons pass through the solid without being absorbed!
- *Conclusion*

: Material is **completely transparent and colorless** (e.g., Diamond  $E_g = 5.5$  eV, Quartz glass  $E_g = 9.0$  eV, Sodium chloride  $E_g = 7.3$  eV).

#### 2. Narrow Bandgap Semiconductors ( $E_g < 1.8$ eV):

- Every visible photon has energy  $h\nu > E_g$ .
- **All visible light is absorbed** within a thin surface layer!
- *Conclusion*

: Material is **completely opaque** and appears black or dark metallic grey (e.g., Silicon  $E_g = 1.1$  eV, Gallium arsenide  $E_g = 1.4$  eV).

#### 3. Intermediate Bandgap Materials ( $1.8 \text{ eV} \leq E_g \leq 3.1 \text{ eV}$ ):

- The material absorbs higher-energy (shorter wavelength) visible photons (violet, blue, green), but transmits lower-energy (longer wavelength) photons (yellow, red).
- *Conclusion*

: Material is **colored and translucent/transparent!**

- *Example*

: Cadmium Sulfide (CdS) has  $E_g = 2.42$  eV ( $\lambda_{\text{cutoff}} = 1.24/2.42 = 0.51 \mu\text{m}$ ). It absorbs blue and violet light ( $\lambda < 0.51 \mu\text{m}$ ) and transmits yellow and red ( $\lambda > 0.51 \mu\text{m}$ ), giving CdS crystals a brilliant, vivid **yellow-orange color!**

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## 5. Comprehensive Step-by-Step Problem Walkthroughs

### 5.1 Problem 1: Electrical Conductivity of Extrinsic n-Type Silicon

**Problem Statement:** An n-type silicon semiconductor is doped with Phosphorus (P) at an impurity concentration of  $N_d = 1.0 \times 10^{17} \text{ atoms/cm}^3 = 1.0 \times 10^{23} \text{ atoms/m}^3$  at room temperature (300 K). Given:

- Pure intrinsic silicon carrier concentration at 300 K:  $n_i = 1.5 \times 10^{10} \text{ cm}^{-3} = 1.5 \times 10^{16} \text{ m}^{-3}$ .
- Electron mobility:  $\mu_e = 0.135 \text{ m}^2/(\text{V} \cdot \text{s})$ .
- Hole mobility:  $\mu_h = 0.048 \text{ m}^2/(\text{V} \cdot \text{s})$ .
- Electron charge:  $e = 1.602 \times 10^{-19} \text{ C}$ .

1. Compute the electrical conductivity of **pure intrinsic silicon**.
2. Compute the electrical conductivity of the **doped n-type silicon**.
3. Determine the factor by which doping increased the electrical conductivity.

### Step 1: Calculate Conductivity of Pure Intrinsic Silicon

$$\begin{aligned}\sigma_{\text{intrinsic}} &= n_i |e| (\mu_e + \mu_h) \\ \sigma_{\text{intrinsic}} &= (1.5 \times 10^{16} \text{ m}^{-3}) \times (1.602 \times 10^{-19} \text{ C}) \times (0.135 + 0.048 \text{ m}^2/(\text{V} \cdot \text{s})) \\ \sigma_{\text{intrinsic}} &= (2.403 \times 10^{-3}) \times (0.183) = 4.40 \times 10^{-4} (\Omega \cdot \text{m})^{-1}\end{aligned}$$

### Step 2: Calculate Conductivity of Doped n-Type Silicon

At room temperature (300 K), all phosphorus donor atoms are completely ionized in the extrinsic regime:

$$n \approx N_d = 1.0 \times 10^{23} \text{ electrons/m}^3$$

Notice that  $n = 10^{23}$  is seven orders of magnitude greater than  $n_i = 10^{16}$ ; hole conduction is completely negligible.

$$\begin{aligned}\sigma_{n\text{-type}} &\approx n |e| \mu_e = (1.0 \times 10^{23} \text{ m}^{-3}) \times (1.602 \times 10^{-19} \text{ C}) \times (0.135 \text{ m}^2/(\text{V} \cdot \text{s})) \\ \sigma_{n\text{-type}} &= 16,020 \times 0.135 = 2,163 (\Omega \cdot \text{m})^{-1} \approx 2.16 \times 10^3 (\Omega \cdot \text{m})^{-1}\end{aligned}$$

### Step 3: Compute the Increase Factor

$$\text{Factor} = \frac{\sigma_{n\text{-type}}}{\sigma_{\text{intrinsic}}} = \frac{2,163 (\Omega \cdot \text{m})^{-1}}{4.40 \times 10^{-4} (\Omega \cdot \text{m})^{-1}} = 4.92 \times 10^6 \approx 5,000,000 \times$$

*Engineering Conclusion*

: Doping silicon with merely **1 phosphorus atom per 500,000 silicon atoms** increased its electrical conductivity by **nearly 5 million times!**

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## 5.2 Problem 2: Optical Bandgap Cutoff & Transparency Prediction

**Problem Statement:** Predict whether the following crystalline materials are optically transparent, colored, or opaque to visible light ( $0.4 \mu\text{m} \leq \lambda \leq 0.7 \mu\text{m}$ ), and compute the cutoff absorption wavelength  $\lambda_{\text{cutoff}}$  (in  $\mu\text{m}$ ) for each:

1. Gallium Phosphide (GaP,  $E_g = 2.26 \text{ eV}$ )
2. Zinc Selenide (ZnSe,  $E_g = 2.70 \text{ eV}$ )
3. Silicon (Si,  $E_g = 1.12 \text{ eV}$ )
4. Diamond (C,  $E_g = 5.50 \text{ eV}$ )

### Calculation Protocol:

$$\lambda_{\text{cutoff}} = \frac{1.24}{E_g (\text{eV})}$$

1. **Gallium Phosphide** (GaP,  $E_g = 2.26$  eV):

$$\lambda_{\text{cutoff}} = \frac{1.24}{2.26} = 0.549 \mu\text{m} = 549 \text{ nm}$$

*Analysis*

: Photons with  $\lambda < 0.549 \mu\text{m}$  (violet, blue, cyan) are absorbed; photons with  $\lambda > 0.549 \mu\text{m}$  (yellow, orange, red) are transmitted.

*Appearance*

: **Translucent / Transparent with a vivid Orange-Red color.**

2. **Zinc Selenide** (ZnSe,  $E_g = 2.70$  eV):

$$\lambda_{\text{cutoff}} = \frac{1.24}{2.70} = 0.459 \mu\text{m} = 459 \text{ nm}$$

*Analysis*

: Absorbs violet and deep blue light ( $\lambda < 459 \text{ nm}$ ); transmits green, yellow, orange, and red.

*Appearance*

: **Transparent Yellow crystal.**

3. **Silicon** (Si,  $E_g = 1.12$  eV):

$$\lambda_{\text{cutoff}} = \frac{1.24}{1.12} = 1.107 \mu\text{m}$$

*Analysis*

: Cutoff wavelength is in the infrared. All visible photons ( $\lambda \leq 0.7 \mu\text{m}$ ) have energies  $h\nu > 1.12$  eV and are completely absorbed.

*Appearance*

: **Completely Opaque** (dark grey metallic sheen).

4. **Diamond** (C,  $E_g = 5.50$  eV):

$$\lambda_{\text{cutoff}} = \frac{1.24}{5.50} = 0.225 \mu\text{m} = 225 \text{ nm}$$

*Analysis*

: Cutoff is deep in the ultraviolet. No visible photons can excite electrons across 5.5 eV.

*Appearance*

: **Completely Transparent and Colorless.**

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## 6. Critical Concordia Exam Traps & Red Flags

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-  **Trap 1: Temperature Effect on Conductivity: Metals vs. Semiconductors:**

- In **metals**, heating **decreases** conductivity ( $\sigma \propto 1/T$ ) due to increased electron-phonon scattering.
- In **intrinsic semiconductors**, heating **exponentially increases** conductivity ( $\sigma \propto e^{-E_g/2kT}$ ) due to exponential carrier generation. Stating that conductivity increases with temperature for a copper wire is a guaranteed loss of marks!
- ⚠ **Trap 2: The Wiedemann-Franz Law Applies ONLY to Metals:**  $\frac{k}{\sigma T} = L$  is valid **only when free electrons conduct both heat and electricity**. It is completely invalid for ceramics and polymers, where heat is carried by phonons and electrical conductivity is zero!
- ⚠ **Trap 3: Wavelength vs. Energy Inverse Relationship:** Remember:  $\lambda = 1.24/E_g$ . Shorter wavelength means **higher photon energy**! Blue light has more energy than red light. A material that absorbs blue light has a larger bandgap than one that absorbs red light.
- ⚠ **Trap 4: Thermal Shock Resistance Formula Terms:** In  $TSR \cong \frac{\sigma_f k}{E \alpha_l}$ ,  $E$  and  $\alpha_l$  are in the **denominator**! Higher elastic modulus or higher thermal expansion **drastically lowers** thermal shock resistance.